How to Enable LLM with 3D Capacity? A Survey of Spatial Reasoning in LLM

Jirong Zha[1*], **Yuxuan Fan**[2*], **Xiao Yang**[2], **Chen Gao**[1†], **Xinlei Chen**[1†]
[1]Tsinghua University
[2]The Hong Kong University of Science and Technology (Guang Zhou)
zhajirong23@mails.tsinghua.edu.cn, {yfan546, xyang856}@connect.hkust-gz.edu.cn,
chgao96@gmail.com, chen.xinlei@sz.tsinghua.edu.cn

Abstract

3D spatial understanding is essential in real-world applications such as robotics, autonomous vehicles, virtual reality, and medical imaging. Recently, Large Language Models (LLMs), having demonstrated remarkable success across various domains, have been leveraged to enhance 3D understanding tasks, showing potential to surpass traditional computer vision methods. In this survey, we present a comprehensive review of methods integrating LLMs with 3D spatial understanding. We propose a taxonomy that categorizes existing methods into three branches: image-based methods deriving 3D understanding from 2D visual data, point cloud-based methods working directly with 3D representations, and hybrid modality-based methods combining multiple data streams. We systematically review representative methods along these categories, covering data representations, architectural modifications, and training strategies that bridge textual and 3D modalities. Finally, we discuss current limitations, including dataset scarcity and computational challenges, while highlighting promising research directions in spatial perception, multi-modal fusion, and real-world applications.

1 Introduction

Large Language Models (LLMs) have evolved from basic neural networks to advanced transformer models like BERT [Kenton and Toutanova, 2019] and GPT [Radford, 2018], originally excelling at language tasks by learning from vast text datasets. Recent advancements, however, have extended these models beyond pure linguistic processing to encompass multimodal ability (In this paper, when we refer to LLMs, we specifically mean those that integrate multimodal functions). Their ability to capture complex patterns and relationships [Chen *et al.*, 2024a] now holds promise for spatial reasoning tasks [Ma *et al.*, 2024b]. By applying these enhanced models to challenges such as understanding 3D object relationships and spatial navigation, we open up new opportunities for advancing fields like robotics, computer vision, and augmented reality [Gao *et al.*, 2024].

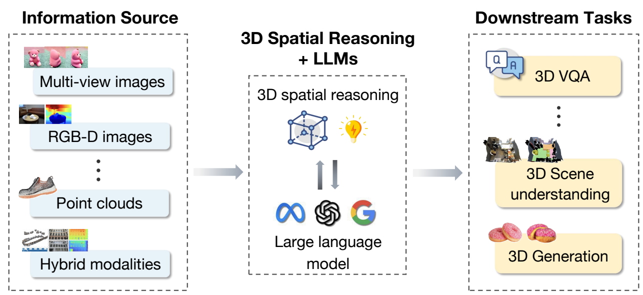

Figure 1: Large Language Models can acquire 3D spatial reasoning capabilities through various input sources including multi-view images, RGB-D images, point clouds, and hybrid modalities, enabling the processing and understanding of three-dimensional information.

At the same time, 3D data and 3D modeling techniques have seen significant developments [Ma *et al.*, 2024c], finding extensive applications in virtual and augmented reality, robotics, autonomous vehicles, gaming, medical imaging, and more. Unlike traditional two-dimensional images, 3D data provides a richer view of objects and environments, capturing essential spatial relationships and geometry. Such information is critical for tasks like scene reconstruction, object manipulation, and autonomous navigation, where merely text-based descriptions or 2D representations may fall short of conveying the necessary depth or spatial context.

LLMs help Spatial Understanding. Bringing these two fields together—powerful language understanding from LLMs and the spatial realism of 3D data—offers the potential for highly capable, context-aware systems. From a linguistic perspective, real-world descriptions often reference physical arrangement, orientation, or manipulations of objects in space. Text alone can be imprecise or ambiguous about size, shape, or relative positioning unless one can integrate a robust spatial or visual understanding. Consequently, there is growing interest in enhancing LLMs with a "3D capacity" that enables them to interpret, reason, and even generate three-dimensional representations based on natural language prompts. Such an integrated approach opens up exciting prospects: robots that can follow language instructions more effectively by grounding their commands in 3D context, architects who quickly prototype 3D layouts from textual descriptions, game designers who generate immersive environments for narrative-based experiences, and many other creative applications yet to be envisioned.

Motivation. Although LLMs have been increasingly applied in 3D-related tasks, and Ma *et al.* [2024b] provided a systematic overview of this field, the rapid advancement of this domain has led to numerous new developments in recent months, necessitating an up-to-date survey that captures these recent breakthroughs. Integrating 3D capacity into LLMs faces several key challenges: (1) the scarcity of high-quality 3D datasets compared to abundant text corpora; (2) the fundamental mismatch between sequential text data and continuous 3D spatial structures, requiring specialized architectural adaptations; and (3) the intensive computational requirements for processing 3D data at scale. While early attempts at combining language and 3D have shown promise, current approaches often remain limited in scope, scalability, and generalization capability. Most existing solutions are domain-specific and lack the broad applicability characteristic of text-based LLMs.

Contribution. The contributions of this work are summarized in the following three aspects: **(1)** *A structured taxonomy*. We provide a timely and comprehensive survey that distinguishes itself from the systematic overview offered by Ma *et al.* [2024b] by presenting a novel perspective on LLM applications in 3D-related tasks: our work constructs a structured taxonomy that categorizes existing research into three primary groups (Figure 2) and offers a forward-looking analysis of the latest breakthroughs, thereby underscoring our unique contributions and the significance of our approach in advancing the field. **(2)** *A comprehensive review*. Building on the proposed taxonomy, we systematically review the current research progress on LLMs for spatial reasoning tasks. **(3)** *Future directions*. We highlight the remaining limitations of existing works and suggest potential directions for future research.

2 Preliminary

2.1 Large Language Models

Large Language Models (LLMs) have evolved from early word embeddings to context-aware models like BERT [Kenton and Toutanova, 2019]. Generative transformers such as GPT series [Radford, 2018], have further advanced text generation and few-shot learning. However, these models often struggle with spatial reasoning due to their focus on textual patterns, prompting efforts to integrate external spatial knowledge [Fu *et al.*, 2024].

Vision-Language Models (VLMs) extend LLMs by aligning visual data with text. Early examples like CLIP [Radford *et al.*, 2021] leverage co-attentional architectures and contrastive learning, while later models such as BLIP [Li *et al.*, 2022] refine these techniques with larger datasets. Yet, most VLMs process only 2D data, limiting their ability to capture detailed 3D spatial configurations. Integrating 3D context via depth maps, point clouds, or voxels remains challenging, motivating ongoing research toward more robust spatial intelligence.

2.2 3D Data Structures

3D data has different structures, which are essential for understanding the three-dimensional world, and common methods include point clouds, voxel grids, polygonal meshes, neural fields, hybrid representations, and 3D Gaussian splatting. Point clouds represent shapes using discrete points, typically denoted as

$$P = \left\{ p_i \in \mathbb{R}^3 \mid i = 1, ..., N \right\},$$

which are storage-efficient but lack surface topology. Voxel grids partition space into uniform cubes, with each voxel $V(i, j, k)$ storing occupancy or distance values, providing detailed structure at the expense of increased memory usage at higher resolutions. Polygonal meshes compactly encode complex geometries through a set of vertices $\{v_i\}$ and faces $\{F_j\}$, though their unstructured and non-differentiable nature poses challenges for integration with neural networks. Neural fields offer an implicit approach by modeling 3D shapes as continuous and differentiable functions, such as

$$f_\theta : \mathbb{R}^3 \to (c, \sigma),$$

which maps spatial coordinates to color c and density σ. Hybrid representations combine these neural fields with traditional volumetric methods (e.g., integrating f_θ with voxel grids) to achieve high-quality, real-time rendering. Meanwhile, 3D Gaussian splatting enhances point clouds by associating each point p_i with a covariance matrix Σ_i and color c_i, efficiently encoding radiance information for rendering. Each method has its unique strengths and trade-offs, making them suitable for different applications in 3D understanding and generation.

2.3 Proposed taxonomy

We propose a taxonomy that classifies 3D-LLM research into three main categories based on input modalities and integration strategies, as shown in Figure 1: Image-based spatial reasoning encompasses approaches that derive 3D understanding from 2D images. This includes multi-view methods that reconstruct 3D scenes, RGB-D images providing explicit depth information, monocular 3D perception inferring depth from single views, and medical imaging applications. While these approaches benefit from readily available image data and existing vision models, they may struggle with occlusions and viewpoint limitations. Point cloud-based spatial reasoning works directly with 3D point cloud data through three alignment strategies: (1) Direct alignment that immediately connects point features with language embeddings, (2) Step-by-step alignment that follows sequential stages to bridge modalities, and (3) Task-specific alignment customized for particular spatial reasoning requirements. These methods maintain geometric fidelity but face challenges in handling unstructured 3D data. Hybrid modality-based spatial reasoning combines multiple data streams through either tightly or loosely coupled architectures. Tightly coupled approaches integrate modalities through shared embeddings or end-to-end training, while loosely coupled methods maintain modular components with defined interfaces between them. This enables leveraging complementary strengths across modalities but increases architectural complexity.

This taxonomy provides a structured framework for understanding the diverse technical approaches in the field while highlighting the distinct challenges and trade-offs each branch

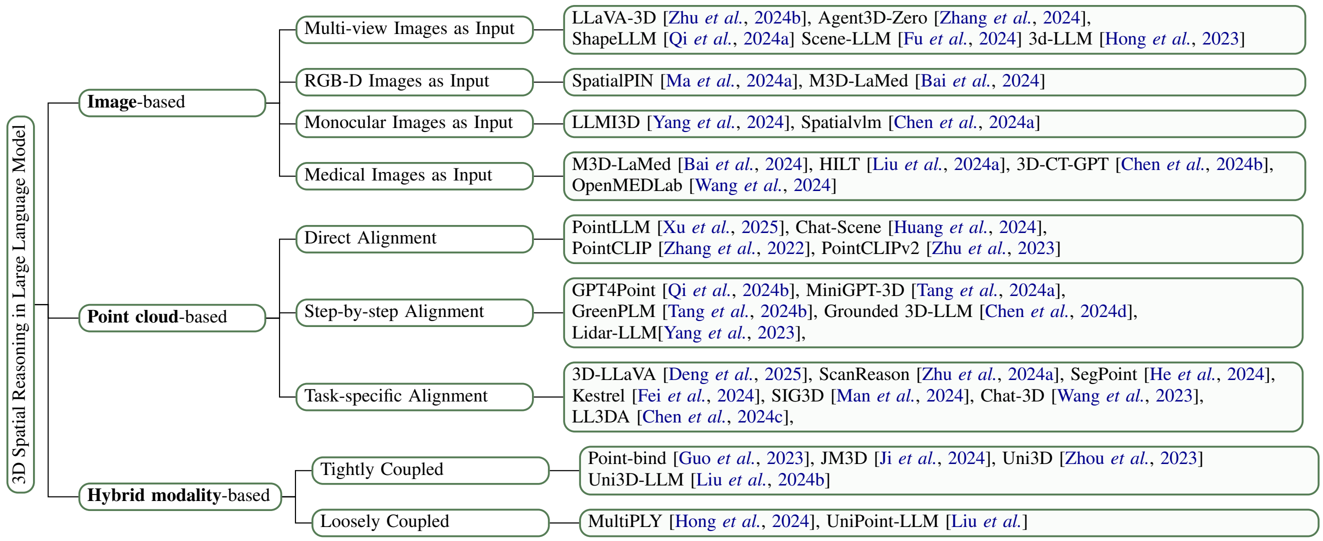

Figure 2: A Taxonomy of Models for Spatial Reasoning with LLMs: Image-based, Point Cloud-based, and Hybrid Modality-based Approaches and Their Subdivisions.

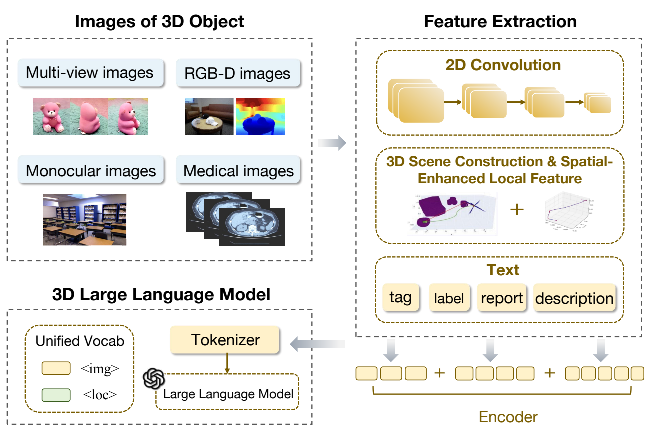

Figure 3: An overview of image-based approaches.

must address. Figure 2 presents a detailed breakdown of representative works in each category.

3 Recent Advances of Spatial Reasoning in LLM

3.1 Image-based Spatial Reasoning

Image-based spatial reasoning methods can be categorized based on their input modalities: multi-view images, monocular images, RGB-D images, and 3D medical images shown in Figure 3. Each modality offers unique advantages for enhancing 3D understanding in Large Language Models (LLMs). Multi-view images provide spatial data from different perspectives, monocular images extract 3D insights from a single view, RGB-D images incorporate depth information, and 3D medical images address domain-specific challenges in healthcare. These categories highlight the strengths and challenges of each approach in improving spatial reasoning capabilities.

3.1.1 Multi-view Images as input

Several studies explore multi-view images to enhance LLMs' spatial understanding. LLaVA-3D Zhu *et al.* [2024b] leverages multi-view images and 3D positional embeddings to create 3D Patches, achieving state-of-the-art 3D spatial understanding while maintaining 2D image understanding capabilities. Agent3D-Zero Zhang *et al.* [2024] utilizes multiple images from different viewpoints, enabling VLMs to perform robust reasoning and understand spatial relationships, achieving zero-shot scene understanding. ShapeLLM Qi *et al.* [2024a] also uses multi-view image input, with robustness to occlusions. Scene-LLM Fu *et al.* [2024] uses multi-view images to build 3D feature representations, incorporating scene-level and egocentric 3D information to support interactive planning. SpatialPIN Ma *et al.* [2024a] enhances VLM's spatial reasoning by decomposing, understanding and reconstructing explicit 3D representations from multi-view images and generalizes to various 3D tasks. LLMI3D Yang *et al.* [2024] extracts spatially enhanced local features from high-resolution images using CNNs and a depth predictor and uses ViT to obtain tokens from low-resolution images. It employs a spatially enhanced cross-branch attention mechanism to effectively mine spatial local features of objects and uses geometric projection to handle. Extracting multi-view features results in huge computational overhead and ignores the essential geometry and depth information. Additionally, plain texts often lead to ambiguities especially in cluttered and complex 3D environmentsChen *et al.* [2024c]. ConceptGraphs Gu *et al.* [2024] proposes a graph-structured representation for 3D scenes that operates with an open vocabulary, which is developed by utilizing 2D foundation models and integrating their outputs into a 3D format through multiview association.

3.1.2 Monocular Image as input

LLMI3D Yang *et al.* [2024] uses a single 2D image for 3D perception, enhancing performance through spatial local feature mining, 3D query token decoding, and geometry-based